

SINKGRAB · DESIGN DECISIONS REGISTER

SinkGrab design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened; design decisions made under Amish's pre-approvals of 2026-10-03; three requirement decisions (R4, R3, R1) proposed, awaiting Amish

SinkGrab design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below set SinkGrab's safety case (overload limits, lowering on a brake that is on when let go, the doors over the shaft, no entry). Each took the conservative option; the evidence that would relax it is in SKG-DDR-001, Table 1. None of the open decisions below changes the safety case.

■ Open decisions

Requirements that are not met or at risk on paper are decided by Amish (2026-10-03: "A simple statement doesn't add value - ensure you are identifying a state and posing it as a clear recommendation for me to decide on"). The full state, options and effects are in `docs/REVIEW.md`, TRL 3, "Decisions for Amish".

#	Decision needed	Options	Recommendation	Affects in the build	Source	Status
1	R4 cycle time: 3.7 min at 10 m against the target, because two people give about 100 W and the hoist takes 111 s	A: a third person at the cranks for the hoist (3.1 min; one long two-hand handle, USD 10, 0.5 kg). B: a smaller grab, 240 mm shells (about 3.4 min; bite 15 L, R3 not met; USD 10 less, 4 kg less). C: keep two people and make 4 min the trial target (3.7 min; no cost or mass change)	A	One crank handle 240 mm long for two hands	SKG-CAL-001, J	Proposed, awaiting Amish
2	R3 bite: 21.2 L at an assumed 75 % fill, 14.1 L at 50 %; the lips close with only 0.71 of the line pull, capped by the grab's 618 N submerged weight	A: keep the 2:1 tackle and settle the fill in test-pit trials (no change). B: 3:1 tackle with a second sheave in the head (lip force about 656 N, +50 %; +1.5 s a cycle; USD 35, 2.5 kg). C: 10 kg more head ballast (lip force about +14 %; working pull 1,114 N, link margin falls to 1.08; USD 10, 10 kg)	B	Head box gains a sheave and axle; line reeved three parts	SKG-CAL-001, B	Proposed, awaiting Amish
3	R1 depth gained: the 8-ring string keeps sinking to 3 m only while skin friction stays below 1.66 kPa; the estimate is 1 to 3 kPa	A: as designed, toes 15 mm past the ring, and measure in the trial (no change). B: longer toes cutting 40 mm past the ring's outer face to loosen the soil against it (friction expected toward the low end; T-bar force +4 %; USD 10, 0.5 kg). C: sixteen saddles in place of eight (limit rises to 1.81 kPa; USD 340 with lines, 165 kg more kit)	B	Toe plates 25 mm longer; centralizer unchanged	SKG-CAL-001, I	Proposed, awaiting Amish

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Release load of the link pins from the coil bought, by breaking three pins	The link must release between 1.20 and 1.47 kN; the pin diameter (about 1.9 mm) is chosen from these breaks	SKG-CAL-001, C
2	Release of the 3 mm crank shear pins from the bar bought	Sized on 240 MPa shear; a stronger bar lets the line pass the tripod's proof load	SKG-CAL-001, C
3	Breaking strength and splice of the 8 mm rope bought	Factors assume 12 kN and 90 % at the splice	SKG-CAL-001, E
4	Sheave groove, bore and width fit the HatchSide cheeks and 20 mm axle	The head sheave is swapped while SinkGrab works	SKG-DDR-002, item 2
5	HatchSide foot plate size and thickness on the tripod as built	The cradle rim and clamp bars are sized on 140 x 280 mm	SKG-DWG-107
6	Brake lining friction on the steel drum	Holding torque assumes 0.35	SKG-CAL-001, F
7	Bearing centre heights and bolt pitches	Pads and top plates are drilled to UCP206 and UCP205 catalogue sizes	SKG-DWG-101
8	Ring size and wall thickness at the trial well	Toe position, skid holes and saddle slot are set for a 1.0 m ring with a 75 mm wall	SKG-DWG-112, 113, 117

■ Value engineering

Value-engineering target: USD 4,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 2,506 (USD 1,494 under the target). Main cost drivers and savings worth trying:

- The four-gas detector (USD 550) is the largest line; a crew or partner that already owns a pumped detector saves it.
- The eight ballast saddles (USD 240) and their lines (USD 100): saddles cast in concrete inside a steel strap would cost less, at more bulk.
- The two grab shells (USD 120) need plate rolls; a fabricator with rolls is the cheapest route, a cut-and-fold faceted skin the fallback.
- The pole sections (USD 108 for six) scale with well depth.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	Two-shell clamshell grab worked on one closing line, opened by the head's weight	Amish: "I pre-approve the batch runs along with any recommendations you come up with." and "Proceed with the remaining 15 scaffolds"	SKG-DDR-001, items 1, 2
2026-10-03	Shear link (1.20 to 1.47 kN) and 3 mm crank shear pin keep the line inside the HatchSide rating and proof load	Amish, as above	SKG-DDR-001, item 3
2026-10-03	SinkGrab hand capstan (single drum, 4:1 chain, pawl, band brake) in place of the HatchSide winch	Amish, as above	SKG-DDR-001, item 4
2026-10-03	Drawbar to a cradle under the leg A foot; no ground anchors	Amish, as above	SKG-DDR-001, item 5
2026-10-03	Lowering on a weighted band brake with the cranks removed	Amish, as above	SKG-DDR-001, item 6
2026-10-03	Centred scraper pole with centralizer and weight-opened arms	Amish, as above	SKG-DDR-001, item 7
2026-10-03	No de-watering; sump at most 330 mm below the cutting edge	Amish, as above	SKG-DDR-001, item 8
2026-10-03	Tilt read every cycle; correct from 1 in 160; stop undercutting at 1 in 80	Amish, as above	SKG-DDR-001, item 9
2026-10-03	Ballast saddles on their own lines in place of a ballast frame	Amish, as above	SKG-DDR-001, item 10
2026-10-03	Well-head frame with folding doors and spoil tubs in place of a chute and tipping hook	Amish, as above	SKG-DDR-001, item 11
2026-10-03	First co-design candidate to approach: Kerala Ground Water Department (not agreed); second, a rural water NGO in the Ethiopian highlands	Amish, as above	SKG-DDR-001, item 12
2026-10-03	HatchSide tripod used with a swapped 8 mm head sheave; SinkGrab capstan recorded as the single-drum variant of the common capstan block; CalRig as first candidate proof-load rig	Amish, as above	SKG-DDR-001, item 13
2026-10-03	Targeted patent search for under-curb scraper claims before public release	Amish, as above	SKG-DDR-001, item 14
2026-10-03	budget_usd kept at 4,000 as a value-engineering target	Amish: "I also accept any cost overruns or variations from the assumed scope cost."	SKG-DDR-001, item 15
2026-10-03	Design for construction: the fifteen changes of SKG-DDR-002	Amish, as for item 1	SKG-DDR-002
2026-10-03	Scraper working limit 25 m (12 pole sections)	Amish, as for item 1	SKG-CAL-001, H

Date	Decision	Decided by	Record
2026-10-03	Appearance model additions for renders: wound rope, context collar, ground, tub, ring pieces and mannequin	Amish, as for item 1	docs/ REVIEW.md, TRL 3